Mohamed Aradaib

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Education

Iowa State University — Ames, IA
Bachelor of Science in Computer Science — *Aug. 2021 – May 2025*

Professional Experience

Quality Assurance Engineer — Feb. 2026 – Present
Henry Schein One — *Ames, IA*
- Built and maintained automated test frameworks, improving software quality and accelerating release cycles.
- Developed test scripts and debugging solutions using programming and CI/CD tools to identify and prevent defects.
- Collaborated with software engineers to validate features, optimize test coverage, and deliver reliable, scalable applications.

Data Center Technician — Oct. 2025 – Feb. 2026
Microsoft — *West Des Moines, IA*
- Analyzed system behavior and service alerts to identify issues and maintain stable infrastructure operations
- Performed hardware installs, diagnostics, and component replacements while documenting issues for engineering teams.
- Supported deployment workflows by validating configurations, reviewing logs, and ensuring smooth rollout of new services.

Teaching Assistant - Data Structures (Java) — Aug. 2024 – Dec. 2024
Iowa State University — *Ames, IA*
- Guided students in implementing and analyzing core data structures such as arrays, linked lists, trees, and graphs.
- Helped students evaluate algorithm efficiency and identify performance bottlenecks using Big-O analysis.
- Led code-review sessions to strengthen correctness, readability, and modular Java design.

Teaching Assistant - Introduction to Computer Programming (Python) — Aug. 2022 – Dec. 2022
Iowa State University — *Ames, IA*
- Assisted students with Python fundamentals, debugging, and problem-solving.
- Led weekly lab sessions and held office hours for individualized support.
- Reviewed programming assignments and contributed to exam grading.

Projects

Rag Document Assistant | *Python, FastAPI, Streamlit, LlamaIndex, Qdrant, Inngest, Docker* — Live Demo | GitHub
- Retrieval-Augmented Generation AI agent that delivers accurate, context-aware answers from documents.
- Fast pipeline for ingestion, embedding, and vector search to ground LLM responses.
- Simple UI and backend flow that lets users query and interact with the agent seamlessly.

LedgerBank | *Java, Spring Boot, Bootstrap, Thymeleaf, MySQL* — Live Demo | GitHub
- Bank application with account creation, deposits, withdrawals, and transaction history.
- Secure authentication and validation to protect user financial data.
- Persistent storage for customer profiles and transactions to ensure reliable data management.

Collaborative IDE | *JavaScript, React, HTML, CSS, Node.js, Express.js, SQLite* — Live Demo | GitHub
- Real-time collaborative code editor with room-based multi user coding and live synchronization.
- Event-driven WebSocket communication syncing code changes, user actions, and editor state across connected clients.
- IDE-like browser experience with multi-language syntax highlighting and responsive React interface.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML, CSS
Frameworks & Libraries: React, Node.js, Express.js, Spring Boot, FastAPI, LlamaIndex, Streamlit, Bootstrap, Thymeleaf
Databases: PostgreSQL, MySQL, SQLite, Qdrant
Developer Tools: Git, GitHub, Docker, Postman

Extracurriculars

National Society of Black Engineers — Aug. 2022 – May 2023
Board Member — *Iowa State University*
- Coordinated chapter-wide technical initiatives to improve member skill development and event quality.
- Secured $10k+ from sponsors to host corporate tech talks and engineering resume reviews
- Managed event timelines and resources to increase overall program effectiveness.

Computer Science and Software Engineering Club — Aug. 2021 – May 2025
Member — *Iowa State University*
- Collaborated on software projects during hackathons and coding events.
- Practiced algorithms and data structures through weekly technical problem-solving sessions.
- Attended club workshops and tech talks to explore new programming tools, frameworks, and industry practices.